7(a)(i)	Given: $\lambda = 4.5 \times 10^{-7}$ m; Intensity = 700 W m^{-2}; Area A = 1.0×10^{-4} m^2 Total energy of the beam of light, $E = \frac{Nhc}{\lambda}$ Intensity = P/A = E /At = $\frac{Nhc}{\lambda At}$ No. of photons per second, $N/t = \frac{\text{Intensity}(A)(\lambda)}{hc}$ $N/t = (700 \text{ W m}^{-2})(4.5 \times 10^{-7} \text{ m})(1.0 \times 10^{-4} \text{ m}^2) / hc$ $N/t = 1.6 \text{ or } 1.58 \times 10^{17}$	[1] [1]
7(a)(ii)	Momentum of each photon, $p = h/\lambda$ $= h/ 4.5 \times 10^{-7}$	[1]

	= 1.5×10^{-27} or 1.47×10^{-27} N s	[1]
7(a)(iii)	Since the photons are all absorbed (ie not reflected), change in p for collision of ONE photon, $\Delta p = p_f - p_i$ $= p - 0$ = Ans in (ii) $= 1.47 \times 10^{-27}$ N s change in p for collisions of ALL the photons in 1 s = $(N/t)(\Delta p)$ = Ans in (i) x Ans in (ii) $= (1.58 \times 10^{17}) \times (1.47 \times 10^{-27})$ $= 2.3 \times 10^{-10}$ N	[1, ecf] [1, ecf]

7(b)(i)	$\Psi(x) = A \sin 1kx$ at ground state when n = 1 Applying boundary condition at x = L, $\Psi(L) = 0$ since electron cannot be found at boundary with infinitely high potential $A \sin 1kL = 0$ $\sin kL = 0$ or $A = 0$ {trivial case} $kL = n\pi$ $k = \frac{\pi}{L}$ where n = 1	[1] [1]
7(b)(ii)	$\lvert\Psi\rvert^2$ is the probability density, or, probability per unit length (of finding the electron at x)	[1]
7(b)(iii)	Since the particle is confined within a well with infinite potential, the probability of finding the particle within the well = 1 ie 100% Substitute ψ: $\int_0^L A^2 \sin^2\left(\frac{1\pi x}{L}\right) dx = 1$ Since $\int_0^L A^2 \sin^2\left(\frac{\pi x}{L}\right) dx = Area\ under\ \lvert\psi(x)\rvert^2 - x\ graph$ 1 = $(\frac{A^2}{2})$ x L { No further working needs to be shown } $\rightarrow A = \sqrt{\frac{2}{L}}$	[1] [1]